Feedback Is All You Need

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Abstract

Can a small controller acquire high capability by repeatedly incorporating feedback? We test a restricted form of this hypothesis rather than assume that model capacity becomes irrelevant. Across 20 task seeds, four model sizes, and seven feedback conditions in each of two changing classification environments, a 161-parameter learner with accurate feedback outperforms a 2,561-parameter frozen learner by 52.91 percentage points on linear rules and 37.32 points on XOR rules. However, when both receive accurate feedback, the larger model exceeds the smaller by 10.93 points on XOR. A linear model remains near chance on XOR despite repeated correction. Corruption, latency, and replay further condition adaptation. These controlled results support feedback as a mechanism of task adaptation while rejecting an interpretation in which capacity, information, and update rules can be ignored. In a second study of four released language models across 24 hidden-command languages, the 0.5B model receives no measured compositional benefit from 16 correct examples: its accuracy remains 11.46%, and its paired advantage over self-generated examples is 0.00 points (95% CI [-1.03,1.03]). Low-rank parameter updates improve held-out single-command performance but do not establish compositional transfer. These findings constrain the tested mechanisms; they do not determine the limits of every possible feedback system or explain the origin of human intelligence.

1 Introduction

The motivating hypothesis is that a compact cognitive core, coupled to an efficient loop of action, outcome observation, correction, and memory, may acquire capabilities disproportionate to its parameter count. Books and other external records suggest an analogy with accumulated human experience. That analogy motivates investigation; it does not establish a mechanism of human intelligence.

The hypothesis must be distinguished from two stronger assertions: that feedback is literally sufficient without representation or inductive bias, and that existing scaling results are wrong. Training already uses feedback through losses, gradients, rewards, and preferences. Scaling-law work measures relationships among loss, capacity, data, and computation [2, 3]; it does not rule out online adaptation. We therefore compare capacity and feedback as interacting resources.

We retain the researcher's chosen title and follow the organizational pattern of *Attention Is All You Need* [1]: background, mechanism, motivation, experimental specification, results, and conclusion. The title is a hypothesis under examination, not an established sufficiency theorem. Chen et al. already used a closely related title in 2023 [4]; we make no priority claim for the phrase or broad feedback-centered framing.

2 Background

Fast adaptation in a recurrent state was studied in RL^2 [5] and in-context reinforcement learning in Algorithm Distillation [6]. Reflexion stores verbal reflections from task feedback [7], while Self-Refine iterates on self-generated critiques [8]. Intrinsic self-correction can also fail [9]. These are different information sources and update mechanisms and should not be conflated.

Test-time compute studies already show that some smaller models can outperform larger models under specific inference allocations [10]. Consequently, a small-versus-large win alone is not a novel account of intelligence. Our current contribution is a reproducible diagnostic comparison, including conditions under which the motivating hypothesis fails. A broader novelty assessment and language-model evidence remain necessary.

3 Agent and Feedback Model

An agent has parameters θ, persistent memory m, and an interaction history h. Its policy selects an action, and an update U incorporates an environmental response f:

$$a_t \sim \pi_{\theta_t}(\cdot \mid o_t, m_t, h_t), \qquad (\theta_{t+1}, m_{t+1}, h_{t+1}) = U(\theta_t, m_t, h_t, o_t, a_t, f_t).$$

An implementation may update parameters, memory, context, or some combination. Its entire system capacity includes external storage and algorithms; a small neural core is not necessarily a small complete system.

Study A instantiates this notation with a binary classifier whose action is a label prediction. Feedback gives the true label after prediction. Inputs are independent of the action, and gradient descent updates the parameters. This is supervised online learning, not active exploration or a new reinforcement learning algorithm. In a binary task, correctness plus the chosen label also determines the target label.

4 What Feedback Can and Cannot Supply

4.1 An information constraint

Let Θ be uniform over $K > 1$ tasks. Assume the complete initial agent state and all side information are independent of Θ. Count every subsequent task-dependent observation as part of the response transcript. If each of n responses has an alphabet of size at most 2^b, then $I(\Theta; F_{1:n} \mid S_0) \leq nb$. Adaptive queries are functions of S_0 and preceding responses, so cannot independently supply missing task information. Fano's inequality gives the exact-task identification bound

$$P_e \geq \max\left\{0, 1 - \frac{nb+1}{\log_2 K}\right\}.$$

Indeed, $H(\Theta \mid F_{1:n}, S_0) \leq 1 + P_e \log_2 K$; subtracting this entropy from $\log_2 K$ yields the result. The bound is about task identification, not every prediction or reward metric. Informative priors or unaccounted observation channels invalidate these assumptions.

For a binary symmetric channel with independent flip probability p, the per-response information is at most $1 - H_2(p)$ bits. At $p = 1/2$, responses carry no information about the true bit. At $p = 1$, inversion is recoverable if the corruption mechanism is known. Learner failure under inverted labels therefore diagnoses a misspecified channel, not destroyed information.

4.2 Representation and transcript constraints

If a successful policy is outside the complete agent's permitted policy class, additional updates within that class cannot express it. External memory or tools may enlarge that class, so this constraint cannot be inferred from neural parameter count alone. In addition, a deterministic updater replaying the same ordered transcript from the same initial state reaches the same final state, by induction. Calling the procedure a loop adds no independent advantage. Active information acquisition or changing observations can matter; these mechanisms require their own tests.

5 Experimental Setup

5.1 Tasks, models, and feedback

Each seed defines an initial and an independently generated post-change rule over eight standard-normal inputs. Linear labels threshold a normalized random projection. XOR labels combine signs of two randomly selected coordinates, with an independent fair complement bit for each rule. Warmup comprises 256 batches of 32 examples; adaptation uses a further 256 batches from the new rule. Evaluation sets of 2,048 examples per rule never enter updates. Separate uniform-input tests preserve each rule and change its input distribution.

The models are logistic regression and one-hidden-layer ReLU networks of widths 16, 64, and 256: 9, 161, 641, and 2,561 parameters. Adam uses learning rate 0.01 and default moments, with fresh moments after warmup. Every arm at the same capacity and seed starts from identical warmed-up parameters and sees the same arriving input stream.

The seven arms are frozen weights, clean labels, 25% flips, 50% flips, inverted labels, clean feedback delayed by 32 batches, and 25% replay. Replay replaces the final eight new examples in every batch with eight warmup examples sampled with replacement, retaining their original labels. It therefore changes both memory use and fresh-example exposure. Delayed feedback has 224 updates at arrival batch 256; a separate flush processes the remaining 32 without new observations.

5.2 Protocol, statistics, and resources

The full grid uses 20 seeds, 4000–4019, and was frozen in repository history before collection. A development run accidentally exposed seeds 1000 and 1001; the seed set was amended before formal collection, without hyperparameter tuning. A subsequent code audit found overlapping PRNG streams in the first full run (seeds 2000–2019). That run is archived with a superseded-inference notice. This version reports a replication on untouched seeds 4000–4019, with streams derived by SHA-256 of environment, family, role, and step or width. A complete-grid collision check passed. The protocol, hyperparameters, and endpoints otherwise remain unchanged. This is repository registration, not external preregistration.

The primary contrast in each family compares width-16 clean to width-256 frozen at arrival batch 256. This deliberately unequal-information comparison tests adaptation, not compute superiority. We report paired seed-level t intervals at 97.5% for each of these two contrasts (Bonferroni familywise 95%). Secondary 95% intervals are descriptive. Seeds, not checkpoints or individual examples, are the replication units; t inference assumes adequate approximation of the seed-mean sampling distribution.

The run used PyTorch 2.8.0 with CUDA 12.6 on one RTX 4090 D and completed 1,120 arms in 109.54 seconds. Code and environment hashes, all 19,200 checkpoint rows, and analysis are retained. For arithmetic-budget comparisons we count $6WN$ multiply-add-related training FLOPs for W matrix weights and N examples. This proxy excludes optimizer, activation, bias, and hardware

overhead. Warmup and adaptation are reported separately; it is not lifetime compute or a measured energy budget.

5.3 Study B: pretrained language models

We compare Qwen2.5-Instruct models with 0.494, 1.544, 3.086, and 7.616 billion parameters. Exact Hugging Face revisions and package versions are archived. Each of 24 environments (seeds 3000–3023) assigns four named commands distinct hidden offsets modulo eight. A command adds its offset to an integer state; compositions apply two commands. The family is disclosed, while each environment's offsets are withheld except in an oracle control. Train/test inputs are disjoint: 16 single-command feedback examples, 16 held-out single commands, and 32 held-out length-two compositions per environment. These translations commute, so the test does not measure command-order sensitivity.

Frozen-model conditions are no examples, 4/8/16 correct examples, 16 corrupted examples, 16 sequential model guesses, and explicit oracle offsets. Corruption replaces an answer with a uniformly chosen wrong digit with probability 0.5; this channel still conveys information. Self-generated examples use each model's constrained predictions conditioned only on its earlier guesses. All histories share the same demonstration template. Query selection is fixed, so this is passive adaptation rather than action-dependent exploration. Giving the final correct transcript offline produces byte-identical prompts; we do not count duplicate calls as independent evidence of an online effect. A hand-coded modular-offset estimator solves every test after the same clean examples, demonstrating identifiability within the supplied algorithmic family.

We select the most likely first assistant token among digits 0–7 in bfloat16, without sampling or additional reasoning tokens. Each digit is one tokenizer token. We also record unrestricted first-token accuracy, which does not measure full-response validity. This narrow interface can underuse a model's generative reasoning ability and is a central limitation, not an intelligence metric.

For the 0.5B model, rank-8 LoRA updates all query/value projection matrices (alpha 16, zero dropout), using 16 example-free training prompts for 32 passes, batches of four, AdamW at 2×10^{-4}, and a next-digit cross-entropy loss. Each environment starts from the same base checkpoint and fresh adapters and optimizer. Tests establish exact base-weight equality after two adapter train/unload cycles. Held-out tests use either no examples or the full correct context. There is no checkpoint selection. This measures persistence on one language, not lifelong retention across unrelated tasks.

The primary Study B endpoint is the paired 0.5B clean-16 minus self-generated accuracy on compositions, across 24 environments, with a 95% t interval and 10,000-resample paired bootstrap sensitivity interval. All other intervals are descriptive. Comparing released model sizes confounds parameter count with their different pretraining histories; no causal scaling-law estimate is claimed. All model calls run locally on the same RTX 4090 D, with full query predictions, prompt hashes, token counts, measured times, and peak memory.

6 Results

6.1 Adaptation and capacity

The small adaptive model exceeds the large frozen model by 52.91 points on linear tasks (97.5% CI [47.62,58.20]) and 37.32 on XOR ([28.85,45.79]). When both receive feedback, small-minus-large is 0.00 points for linear (95% CI [-0.22,0.22]) and -10.93 for XOR ([-15.39,-6.46]). The latter comparison shows a substantial remaining capacity effect under the tested update schedule. Logistic regression remains near chance on XOR despite clean feedback. Exact XOR is outside this fixed

Table 1: Mean post-change held-out accuracy (%), 20 seeds. All adaptive columns receive the same 8,192 clean examples; frozen models receive none.

Parameters	Linear frozen	Linear clean	XOR frozen	XOR clean
9	45.61	90.67	50.12	49.87
161	45.56	98.53	47.64	84.97
641	45.64	98.71	47.69	96.14
2,561	45.62	98.53	47.65	95.90

linear policy class, but observed chance performance also depends on the loss and optimization: a biased linear threshold can exceed chance under a zero-one objective. We do not identify chance as the best attainable linear-classifier accuracy or a limit for all small controllers.

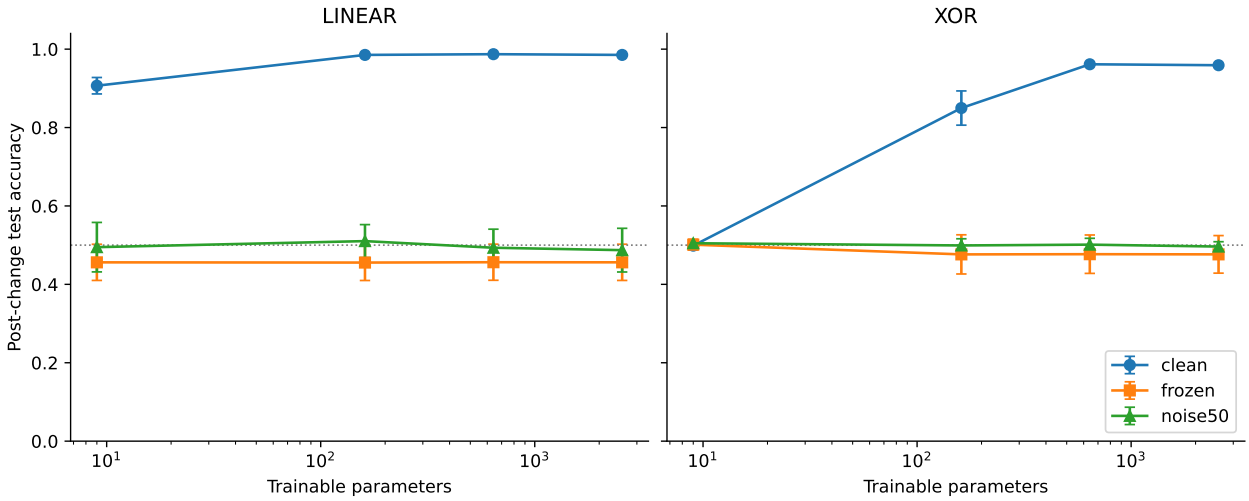


Figure 1: Capacity and feedback jointly determine observed accuracy. Error bars are descriptive 95% t intervals across 20 seeds. The dotted line is chance accuracy; noisy feedback can coexist with information retained from warmup, so individual noisy outcomes need not equal chance.

At an equal adaptation arithmetic proxy, the small model's 256 updates and the largest model's 16 updates both cost 7,077,888 proxy FLOPs. The small model leads by 42.80 points on linear tasks and 34.72 on XOR. However, it receives sixteen times as many new examples, while the large model's warmup cost is sixteen times greater. This is a particular compute/information trade, not an isolated causal effect of feedback-loop efficiency or a scaling law.

6.2 Feedback variations, transfer, and retention

For the 161-parameter model, clean, 25% corruption, and 50% corruption yield 98.53%, 92.05%, and 51.03% accuracy on linear tasks and 84.97%, 71.84%, and 49.92% on XOR. Inverted feedback yields 1.20% and 13.32% respectively. More updates can reinforce the wrong target when the update rule assumes trusted labels.

Delayed feedback reaches 98.28% and 81.87% at the online endpoint. After flushing, its scores and final training losses equal clean feedback for every seed and capacity. This equality is a correctness check for ordered updates, not an additional empirical discovery about latency.

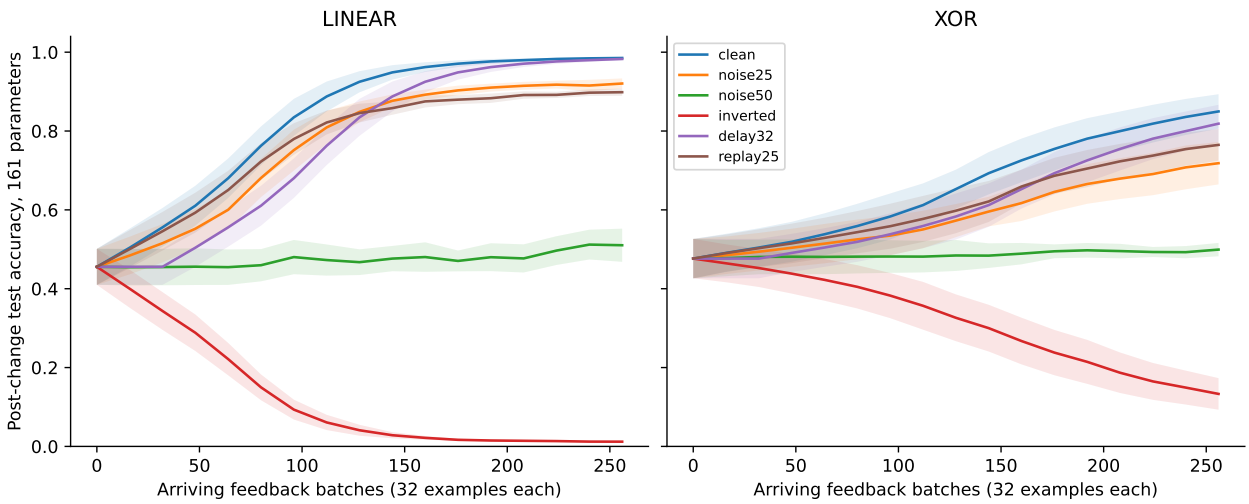


Figure 2: The same 161-parameter learner under six update conditions. Shading shows descriptive 95% seed-level intervals at fixed arrival checkpoints.

Clean small-model uniform-input accuracy is 98.44% for linear rules and 88.19% for XOR. This is distribution transfer within the same known rule family, not transfer to an unrelated task. Old-rule accuracy after clean adaptation is 45.23% and 51.94%. Replay raises these to 54.92% and 64.43% while lowering new-rule accuracy to 89.85% and 76.48%. Because rules may conflict without a task identifier, this is an adaptation– retention tradeoff; perfect performance on both is not always possible.

6.3 Study B: feedback does not guarantee compositional gains

Table 2: Study B mean constrained first-token accuracy (%), 24 environments. Direct tests use one command; composition tests use two. Chance is 12.5%.

Model	Direct none	Direct clean	Compose none	Compose clean	Compose oracle
0.5B	13.80	12.24	11.46	11.46	11.46
1.5B	13.28	13.02	11.20	12.63	12.89
3B	10.16	5.73	13.02	12.63	14.32
7B	13.80	26.56	13.80	17.58	26.82

For the primary 0.5B comparison, clean feedback and self-generated examples both yield 11.46% composition accuracy. The paired difference is 0.00 points (95% t CI [-1.03,1.03]; bootstrap [-0.78,1.04]). Clean-minus-none is also 0.00 points. This is a failure to demonstrate the hypothesized gain with this mechanism, not proof that every feedback mechanism must fail.

The 7B model reaches 26.56% on direct tasks with clean-16 examples versus 13.80% without examples. Its composition scores are 17.58% and 13.80%, respectively: a descriptive paired difference of 3.78 points (95% CI [0.38,7.17]). With explicit offsets, direct accuracy rises to 75.52%, while composition reaches 26.82%. Representation of the instruction, inference of the latent rule, and execution of compositions are distinct bottlenecks.

LoRA trains 540,672 parameters in addition to the frozen 0.494B base. Without example context, its direct accuracy is 26.30% versus 13.80% for the base. The paired direct gain is 12.50 points

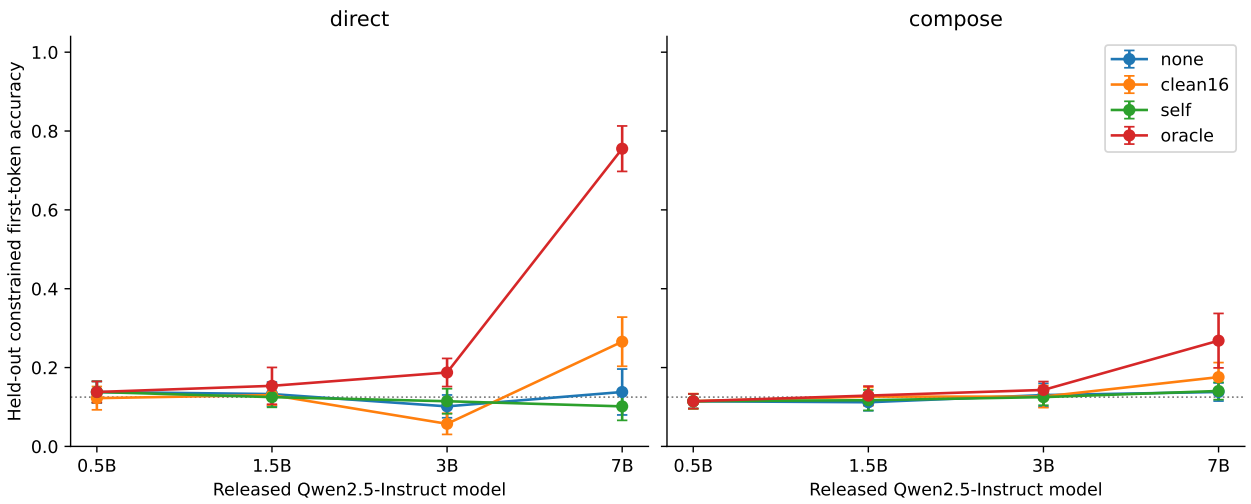


Figure 3: Released-model size and the supplied evidence affect direct and compositional performance differently. Error bars are descriptive environment- level 95% t intervals. The oracle provides extra structured information; it is not a cost-matched competitor.

(descriptive 95% t CI [3.76,21.24]). Composition accuracy is 13.15%; its paired improvement over no-feedback base is 1.69 points (95% CI [-2.12,5.50]). Retaining examples at inference yields 13.28% composition accuracy but reduces direct accuracy to 9.90%. Thus the persistent single-command gain is strongly context-dependent: adding the same demonstrations does not preserve it. These results distinguish a gain under the training prompt format from robust compositional transfer.

Each frozen-model suite processes 2,156,304 unpadded input tokens in 888 forward calls, including self-history construction and evaluation. Measured suite times are 21.81, 56.22, 109.02, and 234.69 seconds; peak allocated memory is 1.20, 3.41, 6.41, and 15.29 GiB. These figures are whole-suite accounting, not equal lifetime budgets or per-condition causal estimates. The LoRA suite adds 1,486,848 training input tokens and 86.67 measured update seconds. Its extra optimization is explicitly not cost-matched to frozen inference.

6.4 Limitations and interpretation

Both studies use synthetic tasks with disclosed families and inexpensive feedback. They do not measure autonomous exploration, long-horizon reward, human intelligence, or general cross-domain skill accumulation. Study A's fixed optimization schedule may interact with capacity. Its replay treatment also changes fresh-example exposure. Study B uses only one model family and a first-token response interface; low accuracy can reflect that interface, insufficient prior execution ability, or the specific update schedule.

In development, 0.5B gave a surprising wrong response to a simple arithmetic calibration. The official weight SHA-256 matched, and CPU float32, GPU float32/bfloat16, and eager/SDPA paths reproduced it. These diagnostics exclude some implementation failures but do not establish that the prompt or scoring interface makes full use of model capability. A prospectively specified full-generation follow-up is necessary before treating these negative results as broadly representative of language agents.

The initial Study A random-stream defect and the corrected fresh-seed replication are both disclosed and preserved. A disk-full error occurred only during Study B export; every original run had completion metadata and complete predictions. Export was repeated on a separate filesystem

without rerunning models. Source hashes and complete outcome grids were checked before analysis.

Prior work already establishes reinforcement learning, external memory, verbal reflection, and test-time computation as relevant mechanisms. Our current evidence is a diagnostic investigation of the stated hypothesis, not a first discovery of feedback-driven intelligence. A targeted literature review, especially its full-text coverage, remains incomplete.

7 Conclusion

Feedback enables substantial adaptation in small neural learners under task change, but capacity remains consequential when both small and large models learn. In our pretrained-language-model protocol, merely retaining correct examples does not yield the hypothesized compositional improvement in the smallest model. A short parameter-update schedule produces a persistent direct skill gain without clear compositional transfer. These findings support studying feedback quality, representation, and update mechanisms jointly. They do not establish that feedback alone explains intelligence, or that increasing model capacity is an incorrect research direction.

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